- Compliant with NIST FIPS publication 197 “Advanced encryption standard (AES)” (November 2001)
- Encryption and decryption with multiple chaining modes:
  - Electronic codebook (ECB) mode
  - Cipher block chaining (CBC) mode
  - Counter (CTR) mode
  - Galois counter mode (GCM)
  - Galois message authentication code (GMAC) mode
  - Counter with CBC-MAC (CCM) mode
- Protection against side-channel attacks (SCA), including differential power analysis (DPA), certified SESIP, and PSA security assurance level 3
- 128-bit data block processing, supporting cipher key lengths of 128-bit and 256-bit
  - 480 or 680 clock cycle latency in ECB mode for processing one 128-bit block with, respectively, 128-bit or 256-bit key
- Hardware secret key encryption/decryption (wrapped-key mode)
- Using a dedicated key bus, optional key sharing with a faster AES peripheral (shared-key mode), controlled by SAES
- Integrated key scheduler to compute the last round key for ECB/CBC decryption
- 256-bit of write-only registers for storing cryptographic keys (eight 32-bit registers)
  - Optional 128-bit or 256-bit hardware loading of two hardware secret keys (BHK, DHUK) that can be XOR-ed together
- Security context enforcement for keys
- 128-bit of registers for storing initialization vectors (four 32-bit registers)
- 32-bit buffer for data input and output
- Automatic data flow control supporting two direct memory access (DMA) channels, one for incoming data, one for processed data. Only single transfers are supported.
- Data-swapping logic to support 1-, 8-, 16-, or 32-bit data
- AMBA AHB slave peripheral, accessible through 32-bit word single accesses only. Other access types generate an AHB error, and other than 32-bit writes may corrupt the register content.
- Possibility for software (in CPU mode only, not in DMA mode) to suspend a message if AES needs to process another message with a higher priority, then resume the original message.

Table 6. AES versus SAES features

Modes or features[1]	AES	SAES
ECB, CBC chaining	X	X
CTR, CCM, GCM chaining	X	X
AES 128-bit ECB encryption in cycles	51	480
DHUK and BHK key selection	-	X
Resistant to side-channel attacks	-	X
Shared key between SAES and AES	X	
Key sizes in bits	128/256	128/256

1. *X = supported.*

3.24 HASH processor (HASH)

The hash processor is a fully compliant implementation of the secure hash algorithm (SHA-1, SHA-2 family) and the HMAC (keyed-hash message authentication code) algorithm. HMAC is suitable for applications requiring message authentication.

The hash processor computes FIPS (Federal Information Processing Standards) approved digests of lengths of 160, 224, and 256 bits for messages of any length less than 2 × 64 bits (for SHA-1, SHA-224, and SHA-256) or less than 2 × 128 bits (for SHA-384, SHA-512).

The HASH main features are: